

Telecom Customer Churn Analysis

Comprehensive Business Intelligence Report

Author: Utkarsh Dhangar

Tools: Microsoft Excel 2019, Pivot Tables, Pivot Charts, Slicers, Git & GitHub

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1. Executive Summary

This report analyzes customer churn behavior in a telecom company using Excel-based business intelligence techniques. The analysis identified customer service interactions, international plans, and heavy daytime usage as the strongest churn drivers. Financial analysis estimated monthly revenue losses exceeding ₹35,000 and lifetime value losses exceeding ₹35 lakh.

2. Business Problem

Telecom companies operate in highly competitive markets where retaining existing customers is more profitable than acquiring new ones. Understanding churn drivers allows businesses to improve retention programs and increase profitability.

3. Dataset Overview

The dataset contains 4,250 customers and 20 variables describing customer demographics, service subscriptions, usage behavior, and churn outcomes.

Dataset Summary

- Total Customers: 4,250
- Churned Customers: 598
- Active Customers: 3,652
- Churn Rate: 14.1%
- Retention Rate: 85.9%
- Missing Values: 0
- Duplicate Records: 0

4. Data Preparation and Cleaning

The dataset was converted into a structured Excel table. Missing values and duplicates were checked and none were found. An audit sheet documented every transformation and validation step to ensure transparency and reproducibility.

5. Basic Analysis Findings

- New Jersey recorded the highest churn rate (27.08%), while Virginia had the lowest (5.00%).
- International plan subscribers experienced churn of 42.17% compared to 11.18% for non-subscribers.
- Voice mail subscribers churned at 7.37% versus 16.44% for customers without a voice mail plan.
- Churned customers made more customer service calls and had higher daytime usage and charges.

6. Intermediate Analysis Findings

- Customers with 4–5 service calls had churn rates of 48.62%.
- Customers with 6+ service calls had churn rates of 64.44%.
- Very high daytime users had churn rates of 48.94%.
- Heavy domestic users represented the most vulnerable behavioral segment with churn of 23.61%.

7. Advanced Analysis Findings

A rule-based risk model and interaction analysis were developed. International plan customers with high international charges demonstrated especially high churn risk. Multiple risk factors combined created significantly higher churn probability than any single variable.

8. Customer Segmentation

Balanced Users (8.90%), International Users (11.81%), Light Users (14.53%), and Heavy Domestic Users (23.61%) were identified through usage-based segmentation. These segments enable targeted marketing and retention campaigns.

9. Risk Model

Low Risk customers showed churn rates of 5.37%, while Medium Risk and High Risk customers exhibited churn rates above 53%. The model can be used to prioritize retention resources.

10. Economic Impact Analysis

ARPU was estimated at ₹59.45 while customer lifetime value reached ₹5,959.37. Churn currently results in monthly revenue losses of ₹35,553.04 and total lifetime value losses of ₹35,63,702.63.

11. Dashboard Overview

A one-page executive dashboard was developed featuring KPI cards, churn drivers, customer segmentation, risk categories, and financial metrics. Interactive slicers enable management to explore customer groups quickly.

12. Business Recommendations

1. Improve customer service resolution rates.
2. Review international pricing plans.
3. Introduce offers for high-usage customers.

4. Launch proactive campaigns for medium and high-risk customers.
5. Monitor customer service interactions as an early warning indicator.

13. Limitations

The dataset lacked temporal variables such as monthly billing dates and churn timestamps, making trend analysis and forecasting impossible.

14. Conclusion

The project demonstrated that Excel can successfully deliver advanced business analytics and actionable retention insights without relying on machine learning techniques.

15. Appendix

Key Metrics:

- Average Revenue Per User: ₹59.45
- Highest State Churn: NJ (27.08%)
- Lowest State Churn: VA (5.00%)
- International Plan Churn: 42.17%
- Voice Mail Plan Churn: 7.37%
- Monthly Revenue Loss: ₹35,553.04
- Total CLV Loss: ₹35,63,702.63